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RESEARCH

One size does not fit all: Why do some new ventures benefit from participating in accelerators more than others?

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Abstract This study investigates why some new ventures derive greater benefits from participation in accelerator programs than others. Specifically, we examine how differences in pre-entry core product knowledge shape ventures' ability to leverage accelerator-based structured learning and mentoring, and how these mechanisms jointly influence venture performance at program graduation (Demo Day) and survival up to 2 years later. We conceptualize learning from the accelerator's structured program through the lens of absorptive capacity (ACAP), encompassing knowledge acquisition, assimilation, transformation, and exploitation. Additionally, we examine mentoring effects through the intensity of mentor–venture interactions and ventures' intentional pivoting in response to mentors' feedback. The analysis draws on a longitudinal dataset of 90 early-stage ventures across 11 cohorts within a single entrepreneurial ecosystem, with data collected at four stages: program entry, midpoint, graduation, and 2-year post-graduation. The findings show that pre-entry core product knowledge strongly predicts both Demo Day graduation performance and short-term survival. However, new ventures with lower initial product knowledge benefit more from a cohort-based, structured learning program, indicating that the learning gained in the accelerator can compensate for limited prior expertise. Moreover, mentoring intensity enhances outcomes only when ventures translate mentors' advice into intentional pivoting. By integrating initial knowledge endowments, accelerator-based learning, and mentoring dynamics, this study advances accelerator research by highlighting the boundary conditions under which accelerators create value and by identifying pivoting as a key mechanism by which mentoring influences venture performance and short-term survival.

Plain English Summary Startup participation in business accelerators has increased steadily in recent years. This study aims to identify which startups benefit most from these programs. Specifically, we examine startups' prior product knowledge and its influence on the learning gained from the accelerator's structured learning program, their performance on Demo Day, and their survival up to 2 years after Demo Day. We found that startups with low, rather than high, prior product knowledge benefit most from the accelerator's structured cohort-based learning program. We also found that the mentoring program was most beneficial for startups that held frequent meetings with their mentors and pivoted in response to the new knowledge gained, making product improvements. These startups achieved high performance on

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Demo Day and short-term survival rates. These findings can help accelerators screen startups and design relevant programs.

Keywords Accelerator · New ventures · Product knowledge · Absorptive capacity (ACAP) · Learning with mentors · Performance on Demo Day and short-term survival

JEL Classification D22 · D80 · D83 · G24 · L26 · M13 · O31

1 Introduction

New ventures operate in turbulent and uncertain environments in which survival depends on their capacity to learn and adapt rapidly (Kim & Lee, 2016; Posen & Levinthal, 2012; Shane & Venkataraman, 2000; Zahra et al., 2000). As nascent organizations, they typically confront acute constraints in knowledge, resources, and legitimacy (Herrmann et al., 2022). To mitigate these liabilities, many ventures engage with business accelerators, which have become central intermediaries within entrepreneurial ecosystems and are designed to compress and intensify venture development processes (Drori & Wright, 2018; Pauwels et al., 2016).

Business accelerators, also referred to as startup accelerators (Brown et al., 2019) or seed accelerators (Canovas-Saiz et al., 2021; Hochberg, 2016), offer structured, time-bound programs that provide comprehensive learning programs, mentoring, networking, and related forms of support (Chang & Assenova, 2026; Stayton & Mangematin, 2019). A growing body of research has examined their impact, focusing on program design features such as duration, mentorship, training activities, networking events, and Demo Days (Dempwolf et al., 2014). Other studies have analyzed accelerator business models, such as for-profit versus non-profit structures, and their implications for venture growth through investment mechanisms, equity arrangements, and access to capital (Cohen et al., 2019a; Hochberg et al., 2015; Miller & Bound, 2011; Shankar & Shepherd, 2019). Overall, accelerators aim to expedite ventures' progress toward a minimum viable product and a sustainable business model (Cohen, 2013; Stayton & Mangematin, 2019) by combining training, mentorship, networking, and investor exposure within a compressed timeframe (Brown et al., 2019; Hochberg, 2016).

Despite their increasing prominence, empirical evidence regarding accelerator effectiveness remains mixed. Some studies report that accelerator graduates outperform comparable non-participants in securing funding and achieving market entry (Gonzales-Uribe & Leatherbee, 2018; Hallen et al., 2020). Others find no consistent performance advantage (Assenova & Amit, 2024; Avnimelech et al., 2024; Cohen et al., 2019a, 2019b), or even higher failure rates among accelerated ventures (Yu, 2020). For instance, Yu (2020) shows that accelerated ventures tend to close earlier and more frequently than non-accelerated counterparts and, conditional on exit, raise less funding. At the same time, these ventures exhibit more efficient investment allocation, as capital is concentrated in those that ultimately succeed rather than in those that eventually fail. This heterogeneity in outcomes suggests that accelerator participation does not benefit all ventures equally. Recent scholarship has therefore called for greater attention to the sources of variability in accelerator returns (Chowdhury & Audretsch, 2024).

Explaining such variation requires to focus on the interplay between the accelerator design and the new ventures' characteristics. In particular, a venture's pre-entry knowledge base, its technological and product-related expertise, shapes its capacity to recognize, assimilate, and apply new knowledge (Cohen & Levinthal, 1990; Zahra & George, 2002). Accelerators function as intensive learning environments that provide opportunities for ventures to acquire, assimilate, transform, and exploit new knowledge, a process known as absorptive capacity (ACAP) (Cohen & Levinthal, 1990; Zahra & George, 2002).

Accelerators vary in their designs, whether for-profit or non-profit, and whether their programs are structured or open-ended (Chang & Assenova, 2026). The extent to which ventures capitalize on these learning opportunities likely depends on the interaction between the type of program and their pre-entry knowledge.

Prior knowledge enhances the ability to recognize, assimilate, and apply new information; however, it may simultaneously engender cognitive rigidity, thereby constraining openness to subsequent learning (Argote et al., 2020; Cohen & Levinthal, 1990). In contrast, ventures possessing limited initial

knowledge may benefit from a compensatory learning effect (Simonsmeier et al., 2022), mainly in structured accelerator programs (Chang & Assenova, 2026). Supporting this perspective, research comparing female and male entrepreneurs finds that prior entrepreneurial experience fully mediates the relationship between gender and outcomes such as entrepreneurial self-efficacy and fundraising capabilities (Avnimelech & Rechter, 2023). Thus, identical accelerator curricula may produce divergent outcomes depending on participants' gender, pre-existing knowledge, and experience.

Beyond cohort-based learning, accelerators provide tailored mentoring, in which experienced entrepreneurs or domain experts offer individualized guidance and problem-solving support (Krishnan et al., 2021; Yitshaki & Drori, 2018). Within iterative learning processes, mentoring may prompt ventures to "pivot," that is, to reorient their product or business model in response to new insights or market feedback (Burnell et al., 2023; Kirtley & O'Mahony, 2023; Ries, 2011). Such pivots represent the transformation and realization of acquired knowledge, consistent with the notion of realized absorptive capacity (Zahra & George, 2002). However, as with cohort-based learning, the effectiveness of mentoring depends on the extent to which ventures internalize and act on the feedback they receive.

Taken together, accelerators provide two complementary learning mechanisms: a structured, cohort-based program that exposes ventures to diverse entrepreneurial knowledge and practices, and a tailored mentoring program that facilitates individualized adaptation. Both mechanisms emphasize ventures' capacity and motivation to learn. Accordingly, the accelerator's impact on new ventures' learning gain and performance is unlikely to be uniform; rather, it is shaped by the interaction among the type of accelerator's learning program, whether structured or open-ended, the new venture's pre-entry knowledge, and the realization of knowledge through intentional action.

To examine these dynamics, we analyze 90 early-stage ventures that participated in a national, multi-branch, multidisciplinary, non-funding, non-profit, and structured accelerator program. Data were collected at four time points: program entry, midpoint, graduation (Demo Day), and up to 2 years of post-accelerator, allowing us to trace the interplay between prior knowledge, learning through ACAP, mentoring intensity, pivoting behavior, and subsequent outcomes in terms of performance on Demo Day and short-term survival.

This study makes three primary contributions. First, it demonstrates that pre-entry core product knowledge is a central determinant of success at the end of the program's Demo Day and of short-term survival of 1- to 2-year post-accelerators. Second, it identifies a compensatory learning effect, demonstrating that new ventures with low, rather than high, initial knowledge derive greater benefits from cohort-based, structured learning programs, as reflected in higher levels of ACAP, knowledge acquisition, and utilization, which translate into improved outcomes, as shown on Demo Day. On the other hand, new ventures with high pre-entry knowledge gained little additional knowledge from the cohort-based, structured program and instead leveraged their prior knowledge to achieve high levels of performance on Demo Day. Third, it shows that mentoring effectiveness depends on the extent to which learning is realized. Ventures that engaged intensively with mentors and translated feedback into intentional, goal-directed pivots achieved higher performance on Demo Day and short-term survival rates.

This study advances the theory and practice of new ventures in accelerators by using the ACAP lenses to examine how accelerators' program characteristics and new ventures' prior knowledge interact to affect new venture short-term success.

2 Theoretical background and hypotheses

2.1 Business accelerator design

Business accelerators are structured programs that foster the rapid development of early-stage ventures through concentrated learning, mentoring, and networking opportunities (Adomdza, 2016; Brown et al., 2019; Cohen, 2013). They typically operate as short-term, cohort-based interventions lasting 3 to 6 months and culminating in a Demo Day: a public pitch event for investors and judges (Dempwolf et al., 2014; Crișan et al., 2021). Although all aim to accelerate the development and growth of the new ventures, they differ in their equity models, program types, and criteria for admitting new ventures,

which may lead to variations in their effects on new ventures' success. Furthermore, even when new ventures attend similar curricular design, outcomes vary widely across ventures (Blair & Shaver, 2019; Cohen & Hochberg, 2014; Hochberg et al., 2015; Cohen et al., 2019b). These variations suggest that the impact of accelerators cannot be fully explained by program structure alone. Instead, it depends on venture-level characteristics, including pre-entry knowledge, learning and development processes, and adaptive behavior towards high performance upon graduation and short-term survival (Frese & Gielnik, 2023). Recently, Chang and Assenova (2026) distinguished between two types of accelerator programs: structured and open-ended, suggesting that structured programs are better suited to new ventures with low prior knowledge, whereas open-ended programs are better suited to educated and knowledgeable founders. Adopting a process perspective, we examine how the accelerator's program structure and the new ventures' pre-entry knowledge jointly shape venture learning gain, performance, and survival.

2.2 Pre-entry knowledge as a foundation for venture success

Pre-entry knowledge, defined as the stock of relevant expertise and experience a venture possesses before entering an accelerator, is a critical antecedent of performance (Cao & Posen, 2022; Hashai & Zahra, 2022; Shepherd & DeTienne, 2005; Shepherd et al., 2021; Simonsmeier et al., 2022). This knowledge helps ventures recognize opportunities, build prototypes, and adapt quickly (Baron, 2007; Cohen & Levinthal, 1989, 1990; Danneels, 2007; Gruber et al., 2024).

In the accelerator context, where ventures must demonstrate rapid progress, core product knowledge is particularly vital. It allows ventures to refine their minimum viable product (MVP) and articulate a clear value proposition at Demo Day (Gruber et al., 2013; Santamaria & Breschi, 2025; Shrader & Siegel, 2007; Yang et al., 2017). We hypothesize:

> H1. Pre-entry core product knowledgeispositive-lyrelatedto new ventures' high performance on accelerator Demo Day.

2.3 Accelerators as learning mechanisms

Accelerators create a controlled environment for intensive knowledge gain. They combine two learning mechanisms:

1. Cohort-based learning, which provides a standardized structure of knowledge across ventures (Cohen et al., 2019a; McDonald & Eisenhardt, 2020). Alternatively, it provides an open-ended structure that enables new ventures to focus on what suits them best (Chang & Assenova, 2026).
2. Mentor-based learning, which delivers customized feedback and problem-solving support tailored to each venture's needs (Krishnan et al., 2021; Yitshaki & Drori, 2018).

> Together, these mechanisms allow ventures to acquire both general entrepreneurial knowledge and specific insights relevant to their business models.

2.3.1 Cohort-based learning and contingency effects

Entrepreneurial success in new ventures is fundamentally rooted in continuous learning (Minniti & Bygrave, 2001). Cohen and Levinthal's (1990) concept of absorptive capacity (ACAP) provides a foundational framework for understanding organizational learning as the process of acquiring, assimilating, transforming, and exploiting knowledge to enhance action and performance (March, 1991; Argote, 2013). They define ACAP as "the ability of a firm to recognize the value of new, external information, assimilate it, and apply it to commercial ends" (Cohen & Levinthal, 1990, p. 128). Extending this framework, Zahra and George (2002) distinguish between two phases of absorptive capacity: potential absorptive capacity (PACAP), which encompasses knowledge acquisition and assimilation, and realized absorptive capacity (RACAP), which involves the transformation and exploitation of knowledge for organizational benefit. In the present study, ACAP serves as a lens for assessing new ventures' learning processes, particularly their capacity to acquire and utilize new knowledge within accelerator programs.

Cohen and Levinthal's (1990) original formulation emphasizes the cumulative nature of learning,

suggesting that absorptive capacity develops over time and may be especially relevant to large, R&D-intensive organizations. More recent research, however, has extended the application of ACAP to new ventures (Flechas Chaparro et al., 2021; Vincent & Zakaryia, 2021), thereby broadening its relevance beyond established firms. This extension is particularly pertinent in accelerator settings, which are characterized by intensive knowledge inflows and structured learning opportunities. Within such environments, ventures are expected to acquire, assimilate, transform, and exploit new knowledge to accelerate their development. ACAP thus constitutes a meaningful mechanism for understanding how new ventures benefit from participation in accelerators.

ACAP theory posits that individuals and organizations with greater prior knowledge possess structural advantages in processing and integrating new information because prior knowledge creates richer cognitive schemas and more developed conceptual frameworks that facilitate learning (Cohen & Levinthal, 1990). Accordingly, a central assumption in organizational learning is that "those who know more can learn more." However, Cohen and Levinthal (1990) also identify a countervailing dynamic, arguing that organizations with substantial prior expertise may require less additional learning to achieve a given level of performance and may experience "lockout" effects (p. 137), whereby reliance on existing knowledge generates resistance to external ideas and a "not invented here" orientation. This tension may be particularly salient in accelerator programs, which are intentionally designed to expose new ventures to intensive learning opportunities through mentorship, peer interaction, customer feedback, and structured educational interventions (Cohen et al., 2019b; Pauwels et al., 2016). Founders do not enter accelerators with equal levels of prior knowledge regarding their ventures' core products, technologies, or markets, and these differences may shape how ventures engage with accelerator learning mechanisms. New ventures possessing extensive prior core product knowledge, often develop stronger cognitive commitment to their existing assumptions and strategic trajectories, which can lead to cognitive rigidity, confirmation bias, and reduced openness to contradictory external feedback (Tripsas & Gavetti, 2000; Levinthal & March, 1993). In this sense, high prior knowledge may paradoxically constrain absorptive capacity because entrepreneurs become less willing to question underlying assumptions or experiment with alternative approaches. By contrast, ventures with lower levels of prior core product knowledge may approach mentor advice, customer insights, and external feedback with greater epistemic openness because their cognitive schemas are less crystallized. Such openness may enhance the recognition, assimilation, and application of new knowledge, the core dimensions of absorptive capacity (Zahra & George, 2002), and facilitate exploratory learning, strategic adaptation, and pivots during the accelerator process. Nevertheless, in accelerator settings, "lockout" may not necessarily represent a disadvantage. Accelerators are characterized by compressed time horizons, intense milestone pressure, and strong expectations that ventures rapidly produce demonstrable outputs for Demo Day presentations to investors and stakeholders (Cohen et al., 2019b; Hochberg, 2016). Under these conditions, ventures with high levels of prior core product knowledge may derive important advantages precisely because they are less likely to fundamentally reconsider their strategic direction. Rather than allocating scarce cognitive and temporal resources toward reassessing assumptions, these ventures can focus on rapid execution, product refinement, investor readiness, and coherent narrative development. Their conviction regarding the correctness of their product or technological trajectory may therefore facilitate speed, decisiveness, and consistency throughout the accelerator process. In this respect, "lockout" may function as a form of cognitive commitment that is adaptive within short-duration entrepreneurial programs. Founders with substantial prior product knowledge frequently possess clearer technical visions, stronger beliefs about technological feasibility, and more confidence in customer needs, enabling them to minimize strategic distraction and avoid excessive pivoting during the accelerator timeframe. Although experimentation and exploratory learning are generally viewed as beneficial under uncertainty (McGrath, 2001; Ries, 2011), accelerators impose unusually compressed learning cycles in which ventures are expected to demonstrate tangible progress within only a few months. Under such temporal constraints, ventures maintaining stable strategic trajectories may outperform ventures that continuously revise their direction in response to external input. This dynamic aligns closely with the "lockout" dimension of absorptive capacity because

prior knowledge may inhibit not the ability to absorb information, but rather the perceived need to engage with new learning opportunities. In accelerators characterized by rapid iteration, milestone pressure, and compressed timelines, lower levels of prior core product knowledge may enhance long-term learning and adaptability through greater receptivity to external knowledge, whereas higher levels of prior knowledge may create advantageous lockout effects that support focused execution, rapid progress, and Demo Day performance.

Recent theoretical and empirical developments challenge the strictly cumulative view of learning, suggesting a compensatory effect: learners with lower prior knowledge may derive greater benefits from structured instructions than those with higher prior knowledge (Chang & Assenova, 2026; Kalyuga et al., 2003). For more knowledgeable learners, the same guidance may become redundant, increase extraneous cognitive load, and impede deeper processing. What is "new and explanatory" to novices may be "already known and distracting" to experts (Sweller et al., 2011). In this case, novices may gain more from structured programs than experts do, to compensate for their lack or low level of prior knowledge. These findings challenge the "one-size-fits-all" of standardized accelerator programs (Chang & Assenova, 2026).

A recent meta-analysis on the effect of prior knowledge on knowledge gain and performance demonstrated that the correlations between "prior knowledge and post-test knowledge tend to be much higher than the correlation between prior knowledge and knowledge gain" (Simonsmeier et al., 2022, p. 42). One possible explanation for the inconsistent findings regarding the relationship between prior knowledge and knowledge gain is the compensatory effect, which suggests that learners with low prior knowledge expend more effort to acquire new knowledge than those with high prior knowledge (Argote et al., 2020). Furthermore, the meta-analysis suggests that the compensatory effect specifically occurs "for instructions with lower cognitive demands" (Simonsmeier et al., 2022, p. 50).

In the context of new ventures in accelerators, the compensatory effect does not contradict ACAP theory when it occurs within an accelerator that aims to provide the basic knowledge entrepreneurs with low, rather than high, prior knowledge need. High prior knowledge positively and significantly relates to performance. However, in terms of acquiring, transforming, and implementing new knowledge, those with lower prior knowledge may benefit more from the accelerator.

Given that the present non-profit accelerator offers a structured program to develop new ventures during their stay, we expect to find a compensatory effect for ventures with low, rather than high, pre-entry product knowledge. These ventures will be more motivated to leverage the accelerator's learning program to close their initial knowledge gap. Ventures with low, rather than high, pre-entry product knowledge will exhibit greater knowledge gain than ventures with high prior knowledge. Therefore, we hypothesize:

> H2. Knowledge gain, as measured by ACAP, moderates the relationship between pre-entry core product knowledge and ventureperformance on Demo Day, such that ACAP benefits ventures with low rather than high initial knowledge.

2.3.2 Tailored learning through mentorship

Beyond the cohort curriculum, accelerators offer one-on-one mentorship, in which experienced entrepreneurs guide ventures through domain-specific challenges (Bliemel et al., 2021; Giourka et al., 2021; Yitshaki & Drori, 2018). Mentors provide targeted advice on product refinement, market positioning, and resource allocation, translating general principles into actionable strategies (Kunaka & Moos, 2019; McKevitt & Marshall, 2015; Rechter & Avnimelech, 2025). Mentorship is thus a personalized learning process that deepens absorptive capacity by facilitating iterative experimentation and reflection (Shepherd & Patzelt, 2021).

Organizational learning is "a process through which experience performing a task is converted into knowledge, which, in turn, changes the organization and affects its future performance" (Argote et al., 2020; p. 5402). The pivoting process, as part of the learning from mentors, serves to convert and exploit the acquired knowledge. Namely, adjust their strategy, business model, and market opportunities to strengthen their competitive advantage. Intentional pivoting is the outcome of a weighted decision, after deliberation and risk assessment, to address the venture's potential setback (Norbäck et al., 2025), and not

mere "trial and error," hence a calculated, validated action for change. A recent study demonstrated that the clarity of what needs to be done to solve a challenge improves over time through pivoting, ultimately leading to a successful solution (Cromwell & Harvey, 2025).

Pivoting occupies an inherently ambiguous position within entrepreneurship research because it can signify either adaptive, learning-driven strategic change or, instability, uncertainty, and lack of strategic coherence. On one hand, pivoting is often conceptualized as a disciplined response to new information acquired through experimentation, customer interaction, and market feedback (Ries, 2011). From a learning perspective, pivots may reflect the venture's ability to update assumptions, abandon ineffective strategies, and reconfigure resources in response to environmental uncertainty, thereby demonstrating flexibility and adaptive capability (Blank, 2013; McGrath, 1999). Such adaptation aligns closely with theories of absorptive capacity and organizational learning, which emphasize the importance of recognizing and assimilating external knowledge in uncertain environments (Cohen & Levinthal, 1990; Zahra & George, 2002). In highly dynamic entrepreneurial contexts, the willingness to pivot may therefore represent evidence of strategic responsiveness and learning agility rather than failure. On the other hand, frequent or poorly justified pivots may also signal the absence of a coherent strategic vision, insufficient commitment to a clear opportunity, or reactive decision-making driven by uncertainty rather than informed experimentation (Hampel et al., 2020). Because early-stage ventures often operate with limited information and under severe resource constraints, distinguishing productive adaptation from erratic strategic behavior is inherently difficult. Excessive pivoting can generate confusion among stakeholders, reduce organizational focus, and undermine executional consistency, particularly when ventures continuously alter their product-market direction without accumulating sufficient learning from prior iterations (March, 1991). This ambiguity makes the context in which pivoting occurs critically important. Pivoting within structured, guided environments such as accelerators may carry different implications because accelerators provide entrepreneurs with mentorship, feedback mechanisms, accountability structures, and interpretive frameworks that help distinguish evidence-based adaptation from arbitrary change (Cohen et al., 2019b; Pauwels et al., 2016). Mentors can assist founders in evaluating whether strategic changes are grounded in validated learning, customer discovery, or market signals rather than emotional reactions or uncertainty-driven overcorrection. Consequently, guided pivoting may serve as a mechanism for disciplined experimentation, enabling ventures to adapt while preserving strategic coherence and executional focus. In this sense, the value of pivoting may depend less on changing direction itself and more on whether a change emerges from structured learning processes supported by credible external guidance and reflective decision-making mechanisms.

Pivoting as realized learning Zahra and George (2002) distinguish between potential ACAP (PACAP): the acquisition and assimilation of external knowledge, and realized ACAP (RACAP): the transformation and exploitation of that knowledge.

We conceptualize mentor meetings as opportunities for potential learning (PACAP) and pivoting as a behavioral manifestation of realized learning (RACAP). Pivoting involves deliberate, evidence-based strategic shifts in response to mentor feedback or market signals (Kirtley & O'Mahony, 2023; Ries, 2011). Ventures that combine frequent mentor interactions with effective pivoting demonstrate active knowledge exploitation.

We hypothesize:

H3. Pivoting moderates the relationship between the frequency of mentors' meetings and performance on Demo Day, such that meetings with mentors, which reflect new knowledge acquisition and assimilation, affect new ventures' high performance only when followed by intentional pivoting—realizing the acquired, assimilated knowledge.

2.4 From accelerator success on Demo Day to post-graduation short-term survival

Demo Day represents the public culmination of a selective and intensive acceleration process, during which venture quality and founders' execution capabilities are evaluated under external scrutiny. Accelerator research demonstrates that expert judges assess new venture's pitches using criteria such as the

viability of the identified customer need, the strength of the market value proposition, and the development of a minimum viable product (MVP) (Kalgovas, 2015; Leurs & Duggan, 2018). High evaluations at Demo Day therefore serve as an important signal of venture quality and are associated with a greater likelihood of attracting investors and achieving early venture success (Cohen et al., 2019b; Hallen et al., 2020).

One of the most fundamental indicators of early venture success is post-accelerator survival. The relationship between accelerator participation and venture survival appears to be particularly pronounced for early-stage ventures participating in non-profit accelerators, where resource constraints are alleviated primarily through the provision of non-financial support rather than direct funding (Seitz et al., 2026). Venture survival is commonly conceptualized as the inverse of venture failure (Shepherd et al., 2000). A venture is considered to have failed when it ceases to exist as a sustainable economic entity; conversely, survival reflects the venture's ability to continue operating independently over time. Given the high uncertainty characterizing the earliest stages of venture development, survival has become the de facto measure of venture performance in entrepreneurship research (Soto-Simeone et al., 2020). Accordingly, survival has been operationalized using both monthly (Xi et al., 2020) and annual measures, reflecting the substantial liability of newness faced by nascent ventures (Honoré, 2022).

Recent evidence further suggests that superior Demo Day performance, as reflected in judges' evaluations, is positively associated with short-term venture survival through a certification mechanism that signals venture quality to external stakeholders and serves as an early indicator of venture potential (de Rassenfosse & van den Heuvel, 2026; Howell, 2020). Extending this argument, de Rassenfosse and van den Heuvel (2026) show that certification generates persistent long-term benefits for ventures whose quality can be reliably assessed by expert judges, whereas monetary prizes produce only short-term effects and primarily benefit ventures with relatively low operating costs.

Taken together, these findings suggest that stronger Demo Day performance enhances the likelihood of venture survival following accelerator graduation. We therefore hypothesize:

H4a. Performance at the accelerator Demo Day is positively related to short-term post-accelerator survival.

2.4.1 *Pre-entry core product knowledge, performance at Demo Day, and post-graduation survival*

New venture pre-entry core product knowledge allows founders to articulate clearer value propositions, demonstrate credible progress, and respond more effectively to evaluator feedback, thereby increasing the likelihood of strong Demo Day performance. Importantly, these advantages persist beyond the accelerator context. Ventures grounded in deeper domain knowledge are better equipped to refine their offerings, avoid costly strategic errors, and align their products with market needs during the early stages of operation (Delmar & Shane, 2006). Given that survival in the first years is heavily contingent on a venture's ability to establish a viable product and secure stakeholder support under conditions of high uncertainty (Seitz et al., 2026), pre-entry core product knowledge should continue to exert a positive influence on survival, both directly and indirectly through its effect on Demo Day success.

We therefore hypothesize:

H4b: Pre-entry core product knowledge is positively related to new ventures'short-termsurvival, viaperformancein accelerators' Demo Day.

2.4.2 *Moderated mediation of core product knowledge, ACAP, success on the accelerator's Demo Day, and survival*

Accelerators are designed to help founders process information, refine their business models, and develop market-facing narratives (Cohen et al., 2019b). Demo Day performance can thus be interpreted as an observable outcome of this learning process: ventures that articulate clearer value propositions, demonstrate traction, and present coherent strategies are likely to have internalized accelerator knowledge more effectively. Therefore, and in accordance with H2, we expect that performance on Demo Day will mediate the moderating effect of ACAP on the relationship between low, rather than high, prior knowledge and performance on Demo Day. We therefore hypothesize:

H4c: Absorptive capacity moderates the indirect relationship between pre-entry core product knowledge and post-acceleratorshort-termsurvival via performance on Demo Day, such that this moderated mediating effect is stronger for ventures with low, rather than high, prior knowledge.

The post-graduation impact of mentorship Mentorship may yield enduring benefits by equipping ventures to continue learning and adapting after graduation (Blank, 2021; Spigel, 2017). Effective mentor–venture relationships foster commitment, knowledge retention, and sustained strategic agility (Rechter & Avnimelech, 2025; St-Jean & Tremblay, 2011). Ventures that maintain close interactions with mentors and engage in intentional pivoting will not only perform better at Demo Day but also exhibit higher survival rates 1- to 2-year post-graduation.

H5. Intentional pivoting moderated the indirect effect of mentorship intensity on post-accelerator short-term survival via success at graduation, such that this indirect effect is stronger for ventures with high, rather than low, levels of intentional pivoting.

3 Methods

3.1 Accelerator type

The early-stage, non-profit accelerator examined in this study prioritizes support and capability development, distinguishing it from equity-based, funding accelerators that typically target more advanced ventures with existing financial backing (Cohen & Hochberg, 2014; Hochberg, 2016). The governance structures of non-funding accelerators are generally characterized by lower levels of formal control, reduced performance pressure, and a strong developmental orientation (Dempwolf et al., 2014; Kohler, 2016; Pauwels et al., 2016). Their learning dynamics emphasize skill acquisition, iterative experimentation, and reflective practice, often proceeding at a more moderate pace than investment-driven programs. This developmental focus enables deeper cognitive restructuring and more substantive capability building, potentially resulting in slower but more sustainable learning outcomes (Stephens et al., 2025).

The accelerator's design included two main learning components:

1. A cohort-based, structured learning program consisting of lectures, workshops, and meetups with domain experts and entrepreneurs attended by the full cohort (see Appendix Table 4).
2. Tailored-based mentoring, where each venture was paired with a dedicated mentor providing tailored guidance on product development, strategy, and business challenges (Krishnan et al., 2021; Stayton & Mangematin, 2019). Mentorship sessions are self-scheduled by the ventures.

While mentor–venture pairings did not always ensure a perfect fit, ventures were referred to additional mentors as needed. The accelerator operated under Lean Startup and Business Model Canvas methodologies (Blank, 2013; Miles et al., 2017; Osterwalder et al., 2015; Ries, 2011; Shepherd & Gruber, 2021), emphasizing iterative development and market validation. Ventures were required to apply knowledge in between sessions, submit weekly deliverables for feedback, and iteratively refine their solutions (McDonald & Eisenhardt, 2020). The accelerator maintained a venture file, tracking the progress, advancements, and changes of each venture from the start to the end of the program. The program concluded with a Demo Day, during which each venture presented to a panel of judges and accelerator leaders who rated their performance across predefined criteria.

Sample This study examines 90 early-stage new ventures that participated in a multidisciplinary (not industry-specific) accelerator program operating across five branches in Israel. The branches are located in the north, the center, and the south of the country. All branches follow the same program. Collecting data through a unified accelerator program enables in-depth, fine-grained, and comparable analyses of the findings (Chowdhury & Audretsch, 2024). The accelerator is a private, non-profit organization partially supported by governmental grants. It does not take equity or invest in the ventures. Each cohort, capped at 12 ventures, followed a standardized 4-and-a-half-month program based on widely adopted accelerator models (Bliemel et al., 2019; Cohen, 2013). The curriculum combined collective, cohort-based learning with individualized mentorship

and required participants to demonstrate progress at a final Demo Day.

Of the 121 ventures that entered the program, 90 completed it (a 75% completion rate) and were included in the final analysis. Sixty-three percent of the ventures' founders were male, and 37% female. The founders' ages ranged primarily between 18 and 54, with only a small proportion (3%) aged over 54. The ages of 12% of the ventures' founders ranged between 18 and 24, 51% ranged between 25 and 34, 25% ranged between 35 and 44, 9% between 45 and 54, and 3% over 54. Industrial heterogeneity included 34% high-tech (ecommerce, software, AI, IoT, e-education), 25% social (on-line software based social media, and other), 8% biotech and med-tech, 8% services, 7% food-tech, 4% art and design, and 14% other (such as: environment, lifestyle…), regrouped and categorized as five dummy variables, as described later under "control variables" (Agarwal et al., 2004). Of the 31 ventures that did not make it to the end of the accelerator program, 32% were high-tech (e-commerce, AI, software) ventures, 26% were social (online software-based, social media, and other), 9% food-related, 6% agri-related, and 22% other. Thirty-five percent of the founders of the startups whose participation has ceased were female, and 65% were male. The demographics of ventures that did not complete the accelerator program were similar to those of ventures that graduated, in terms of industry sector and founder gender. The percentage of ventures with high prior core product knowledge that did not make it to the end was similar to that of ventures with low prior core product knowledge: 9.4% scored 1, 6.3% scored 1.5, 18.8% scored 2, 9.4% scored 2.5, 25% scored 3, 6.3% scored 3.5, 6.3% scored 4, and 18.8% scored 5. These results suggest that prior core product knowledge did not primarily affect the venture's decision to discontinue participation in the accelerator program. Most ventures were in early development: 62% were founded within 6 months prior to program entry, and 15% within 6–12 months.[1]

3.2 Procedure and data collection

Data was collected between 2019 and 2021 from 11 cohorts across the accelerator's five branches: three cohorts in the main branch and two in each of the other four. Following a four-wave design, data were gathered at four distinct time points to minimize common method bias (Podsakoff et al., 2003) and to trace learning and performance over time (Taggar et al., 2024):

- T1 (Program Start): Collected pre-entry data, including prior knowledge, demographics, and venture characteristics (e.g., founding date, size, industry, preliminary funding).
- T2 (Midpoint, ~2 months): Captured self-reported measures of learning and knowledge application, such as absorptive capacity (new knowledge acquisition, assimilation, transformation, and implementation of new knowledge).
- T3 (Demo Day): Measured venture performance based on evaluations from accelerator heads and external judges.
- T4 (12–24 months post-graduation): Assessed venture survival (binary: operational = 1, closed = 0).

At each time point, one founder, typically the primary participant in the program, completed the questionnaires (Kumar et al., 1993; Urban, 2012; Zahra et al., 2007). Responses were linked across waves using coded identifier to preserve anonymity while ensuring longitudinal matching.

Prior to their participation, participants in the study completed a consent form, assuring their anonymity and their voluntary participation.

3.3 Measures

3.3.1 Dependent variables

Demo Day performance (T3) Venture performance on Demo Day, is based on judges who use standardized evaluation criteria and numerical scores (Gius, 2025). The judges' standardized Demo Day evaluation was assessed using a nine-item, five-point Likert scale (1 = very poor, 5 = very good), adapted from standard evaluation rubrics (Kalgovas, 2015; Leurs & Duggan, 2018). Judges rated each venture on the following criteria: market need, value proposition,

[1] The full dataset of the 90 ventures will be available upon request from the first author.

competitive advantage, revenue model, feasibility, traction, MVP quality, team strength, and presentation clarity.

The mean score across these items constituted the overall performance.

Post-graduation survival (T4) In line with entrepreneurship research showing that exit hazards are highest during the early years following firm entry, we employ 1- to 2-year post-program survival as an indicator of early venture viability. Prior studies consistently show that new-firm hazard rates are steep at young ages and tend to decline as ventures progress beyond the initial post-entry period (Audretsch, 1995; Geroski et al., 2010; Hochberg, 2016; Santarelli & Vivarelli, 2007; van Praag, 2003). Consequently, 2-year survival has been widely adopted in empirical research as a meaningful outcome measure capturing whether ventures successfully overcome early liabilities and remain operational (e.g., Hyytinen et al., 2015). Based on the marketing-based survival measurement of new venture traceability (Soto-Simeone et al. 2020), venture survival was measured via a binary indicator (1 = operational, 0 = not operational) based on verified follow-up data collected 12–24 months after graduation (Audretsch et al., 2021; Dencker et al., 2009; Gonzales-Uribe & Leatherbee, 2018; Hallen et al., 2020; Honoré, 2022). In the current dataset, there were no mergers and acquisitions (M&A) cases.

3.3.2 Independent variables

Pre-entry core product knowledge (T1) Two items rated on a five-point Likert scale (1 = very weak, 5 = very strong) served to assess the venture's technological and production expertise—the foundation of its core product knowledge (Blank & Carmeli, 2021; Dencker et al., 2009; Eesley & Roberts, 2012; Gruber et al., 2012). The two-item index (α=.60) represents the venture's ability to develop and refine its core offering before entering the accelerator. Please see Appendix B.

Mentorship intensity (T2–T3) Captured as the total number of one-on-one meetings between each venture and its assigned mentor during the program (Weinberg & Lankau, 2011). This measure reflects the depth of individualized learning opportunities provided through mentorship.

3.3.3 Moderators

Learning as part of the structured accelerator program assessed by Absorptive Capacity (ACAP) measure: acquisition, assimilation, transformation, and exploitation.

Measured using the 14-item scale by Flatten et al. (2011), adapted to the context of new ventures. Items assessed the venture's ability to acquire, assimilate, transform, and apply external knowledge derived from the accelerator's learning program. Responses were collected on a seven-point Likert scale (1 = strongly disagree, 7 = strongly agree) (see Appendix B).

Example item: "To what extent does your venture use external resources (e.g., accelerator training, mentorship, networks, consultants, databases, journals, market research) to obtain and apply knowledge?" The scale demonstrated high reliability (α=.80). To ensure linguistic validity, the questionnaire was translated and back-translated (Behr, 2018).

Pivoting Defined as substantive changes in the venture's product, market, or business model during the program (Burnell et al., 2023). Pivot frequency ranged from 0 to 5, representing the extent of realized learning and intentional adaptation (Kirtley & O'Mahony, 2023). Pivoting areas were predefined, such as strategy, business model, product, and market (Flechas Chaparro & de Vasconcelos Gomes, 2021). The accelerator across all cohorts reported on each venture's pivot at the end of the program, allowing for standardized and unified reporting as external validation.[2]

3.3.4 Control variables

To isolate the focal effects, we controlled for cohort (branch × year), to account for contextual variation across time and location (Shankar & Shepherd, 2019; Yusubova & Clarysse, 2016).

Venture type: categorized as (1) technology, (2) service, (3) design/art, (4) medicine, and (5) other (Agarwal et al., 2004).

Venture age at entry: coded into 0–12 months, 12–24 months, and over 24 months (Larrañeta et al., 2017; Zahra et al., 2007).

[2] Please see the full research models and hypotheses in Fig. 3

Table 1 Intercorrelations of the study's variables

	Mean	SD	1	2	3	4	5	6	7	8	9	10
1. Product knowledge	3.36	1.12										
2. Mentor Meet	2.4	3.28	0.100									
3. ACAP	4.92	1.02	0.019	0.156								
4. Pivot	2.18	1.27	0.007	0.062	0.121							
5.Performance Demo Day	3.48	0.6	**.294****	0.124	0.120	−0.101						
6. Survival $y=1$	0.57	0.49	**.346****	0.041	0.126	0.036	**.511****					
7. Cohort			0.017	−0.204	−0.040	0.171	−0.021	**.260***				
8. Venture age			0.062	−0.084	−0.046	−0.022	−0.084	0.030	0.071			
9. Venture type			0.032	−0.240	0.115	0.020	0.107	0.185	**.213***	0.159		
10. Founder gender			−0.091	−0.042	−0.017	0.105	0.038	0.134	**.217***	−0.104	0.123	

$^{**}P<0.01$, $^{*}p<0.05$

Gender of founder–masculine, feminine (Avnimelech & Rechter, 2023)

4 Results

4.1 Descriptive statistics

All analyses were conducted using SPSS 28.0 (IBM, New York) and Hayes' PROCESS macro (V4.0) (Hayes, 2018).

Table 1 presents the means, standard deviations (SDs), and intercorrelations among study variables.

Core product pre-entry knowledge (PK) was significantly correlated with both success at graduation (as measured by Demo Day performance; $r=0.294$, $p<.01$) and post-graduation survival ($r=0.346$, $p<.01$). Neither ACAP nor mentorship-related learning showed significant correlations with pre-entry knowledge, suggesting that learning processes operate as moderators rather than direct predictors of performance on Demo Day or short-term survival.

Among control variables, cohort year was significantly associated with survival ($r=0.267$, $p<.05$). An ANOVA indicated that the 2020 cohort exhibited the highest survival rate ($F=2.157$, $p<.05$), and a chi-square test confirmed an association between cohort year and survival ($\chi^2=9.298$, $p<.01$). Other controls (venture location, type, or age) were unrelated to outcomes ($p>.05$), as confirmed by additional CROSSTABS tests.

4.2 Hypothesis testing

4.2.1 H1: Effect of pre-entry knowledge on performance on demo day

A linear regression served to test the effect of pre-entry core product knowledge on performance on Demo Day. Results supported H1: $\beta=0.29$, $p<.01$, indicating that ventures with greater prior knowledge achieved higher performance at graduation.

A logistic regression on post-graduation survival confirmed a similar pattern: ventures with stronger pre-entry knowledge were 1.99 times more likely to survive (OR = 1.989, $p<.01$).

4.2.2 H2: Moderating role of ACAP

To test H2, we used PROCESS Model 1 (Hayes, 2018) with ACAP as the moderator. Results showed a significant interaction between product knowledge and ACAP on performance on Demo Day ($B=-0.10$, $p<.05$). Ventures with low or moderate pre-entry knowledge benefited substantially from high ACAP ($B=0.26$, $p<.001$; $B=0.16$, $p<.01$), whereas ventures with high pre-entry knowledge did not ($B=0.06$, $p=.40$).

ACAP alone did not directly predict performance on Demo Day ($B=0.06$, $p=.29$), underscoring its contingent nature.

These results confirm the compensatory learning effect: structured accelerator learning benefits ventures with lower initial expertise (Fig. 1).

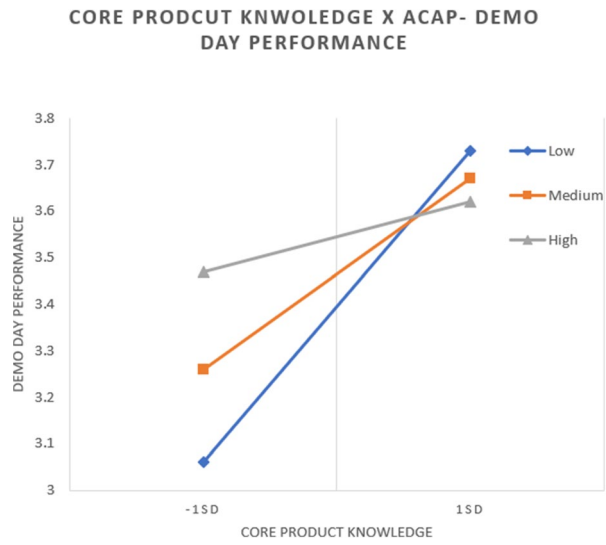


Fig. 1 The relationship between core product pre-entry knowledge and Demo Day performance moderated by ACAP

The interaction between core product knowledge and ACAP shows, that especially those with low core product knowledge benefit most from ACAP (The gray/rectangle symbolized line)

4.2.3 H3: Pivoting as a moderator of learning from mentors

PROCESS Model 1 tested the interaction between mentor meeting intensity and pivoting on performance on Demo Day. Results supported H3: the interaction term was significant ($B = 0.047$, $p < .001$). Mentorship positively affected performance on Demo Day only at high levels of pivoting ($B = 0.13$, $p < .05$), but not at low or moderate levels ($B = -0.009$, $p = .74$; $B = 0.038$, $p = 0.11$, respectively).

This pattern supports the notion that mentorship contributes to venture success primarily when ventures actively translate feedback into intentional action, consistent with realized ACAP (RACAP). Moreover, the pattern shows that high levels of pivoting, combined with a small number of meetings with mentors, lead to poor performance and low success. Consistent with the assumption that pivoting is instrumental when it follows a deliberate, intentional learning process with the mentors (Fig. 2).

4.2.4 H4a. The relationship between success on accelerator's Demo Day and short-term post-graduation short-term survival

The findings from a logistic regression showed a significant effect of new venture performance on Demo Day on the likelihood of new ventures' short-term survival (OR = 8.29, $p < 0.01$). Results support hypothesis 4a.

H4b. The mediating effect of performance in accelerators' Demo Day onthe relationship between pre-entry core product knowledge and post-graduation short-term survival PROCESS Model 4 was used to test hypothesis 4b, which tested the mediating effect of performance on accelerator's Demo Day and the relationship between pre-entry core product knowledge and post-graduation short-term survival. The findings show a significant mediating effect of performance on accelerator's Demo Day on the relationship between core product knowledge and short-term survival ($B = 0.31$, 95% CI [0.087, 0.683]. The finding also shows a significant direct effect of core

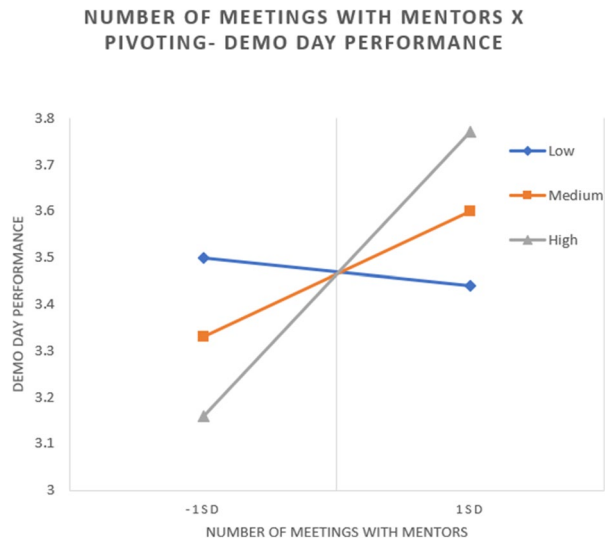


Fig. 2 The relationship between learning meetings with mentors and performance on Demo Day moderated by pivoting

The interaction shows that high number of meetings with mentors and high number of pivots, interact together to perform high on Demo Day. High pivoting by itself, does not relate to high performance on Demo Day. (The gray/rectangle symbolized line)

product knowledge on short-term survival ($B=0.51$, 95% CI [0.028, 0.993]. The confidence intervals for the direct and indirect effect did not include zero; therefore, the findings show both significant direct and indirect effects of mediation. Thus, supporting H4b.

H4c: Moderated mediation via ACAP PROCESS Model 7 was used to test hypothesis 4c (see Table 2 and Fig. 3). H4c tested whether ACAP moderates the indirect relationship between pre-entry knowledge and survival via performance on the Demo Day as the mediator. Results supported H4: The moderated mediation index was significant ($B=-0.21$, 95% CI [−0.59, −0.02]). Pre-entry product knowledge was significantly related to performance on Demo Day ($B=0.61$, 95% CI [0.18, 1.21]). ACAP was significantly associated with performance on Demo Day ($B=0.43$, 95% CI [0.07, 0.78]). Pre-entry product knowledge, moderated by ACAP, significantly predicted performance on Demo Day ($B=-0.108$, $p<.05$), which in turn predicted survival. The confidence interval for the indirect effect did not include zero, indicating significant moderated mediation. Namely, new ventures that compensated for their low levels of prior knowledge by learning and achieving high ACAP scores were successful on Demo Day and, in the short term, survived the post-accelerator period. The direct effect of product knowledge on short-term survival remained significant ($B=0.51$, 95% CI [0.03, 0.99]), indicating partial mediation.

4.2.5 H5: Moderated mediation via pivoting

Finally, PROCESS Model 7 tested whether pivoting moderates the indirect effect of mentor meetings on survival via performance on Demo Day as the mediator (see Fig. 3 and Table 3). The interaction effect between mentorship intensity and pivoting on survival, as mediated by performance on Demo Day, was significant ($B=0.047$, $p<.05$).

The indirect effect of mentorship on short-term survival through performance on the Demo Day was significant under moderate and high pivoting ($B=0.155$, 95% CI [0.03, 0.46]; $B=0.338$, 95% CI [0.08, 0.86]), but not under low pivoting ($B=-0.082$, 95% CI [−0.23, 0.35]).

Robustness checks that included cohort year as a control confirmed that results remained stable (product knowledge → survival: OR = 1.949, $p<.01$; moderated mediation: $B=-0.24$, 95% CI [−0.76, −0.03]). Additionally, gender, as a control variable, did not have a significant effect on the results (product knowledge → survival: OR = 2.146,

Table 2 Performance on Demo Day as a mediator of the indirect relationship between core product knowledge and short-term survival

	B	SE	*t*	*P*	95% CI
		Predicting the mediator Performance on Demo Day			
Core PK	0.69	0.26	2.68	0.009	[0.18, 1.21]
ACAP	0.43	0.18	2.39	0.019	[0.07, 0.78]
PK × ACAP	−0.11	0.05	−2.14	0.036	[−0.21, −0.01]
		Core PK simple slope			
Low	0.26	0.08	3.41	0.001	[0.11, 0.42]
Mid	0.16	0.06	2.71	0.008	[0.04, 0.27]
High	0.05	0.08	0.60	0.551	[−0.11, 0.20]
		Predicting the outcome Performance on Demo Day			
Core PK	0.51	0.25	2.07	0.038	[0.03, 0.99]
Demo Day	1.94	0.52	3.74	< 0.001	[0.92, 2.96]
	Indirect moderating effects for various levels of ACAP				
Low	0.51	0.26	–	–	[0.22, 1.19]
Mid	0.31	0.16	–	–	[0.10, 0.72]
High	0.09	0.17	–	–	[−0.22, 0.47]

PK product knowledge, *ACAP* absorptive capacity, *Demo Day* performance on demo day

$p < .001$; moderated mediation: $B = -0.21$, 95% CI $[-0.62, -0.03]$. These robustness checks align with Simmons et al. (2011) recommendation to report analyses with and without covariates, ensuring transparency and verifying the stability of main findings.

5 Discussion

5.1 Overview of key findings

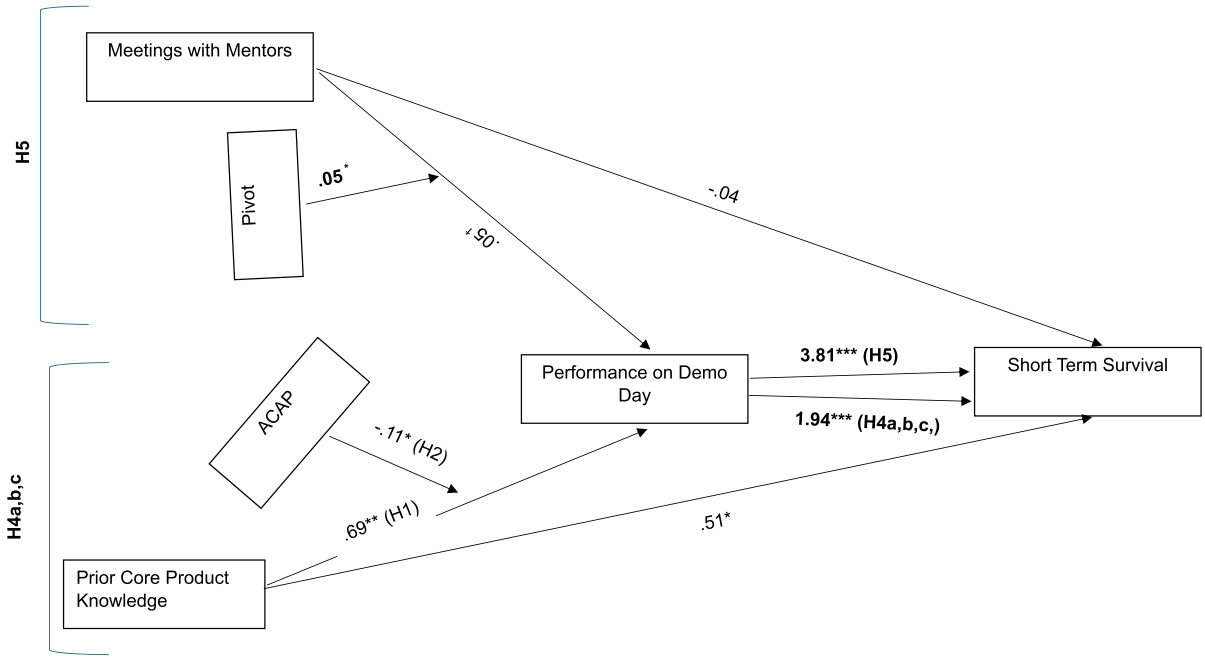
This study provides the answer to the research question "Why do some new ventures benefit from participating in accelerators more than others?" We show that one size does not fit all. Rather, new ventures' pre-entry knowledge interacts with the new ventures' absorptive capacity while attending the non-profit accelerator's structured program to shape the ventures' performance on Demo Day and their 2-year post-accelerator short-term survival. By employing a longitudinal design across four data waves and 90 ventures, we reveal that accelerators operate as selective learning environments, whose benefits are contingent on what ventures know and how they learn.

Three central findings emerged:

1. Pre-entry core product knowledge is a strong predictor of the new ventures' performance on Demo Day, at the end of the program, and their 2-year post-program short-term survival.

2. Learning through ACAP from a cohort-based, structured program exerts a compensatory effect, benefiting ventures with limited initial knowledge more than those with extensive product knowledge and expertise.

The findings of this study reinforce and extend prior research on the relationship between prior knowledge, performance, and learning (Simonsmeier et al., 2022) by demonstrating a distinction between the effect of prior knowledge on performance versus its effect on learning. Consistent with cognitive learning theory and absorptive capacity research, ventures with high prior core product knowledge consistently exhibit superior performance outcomes relative to those with lower prior knowledge because they possess richer cognitive schemas, deeper domain expertise, and more developed conceptual frameworks that support effective execution and decision-making (Cohen & Levinthal, 1990; Simonsmeier et al., 2022). However, the findings also show that lower prior core prod-



Note: † $p < .07$, * $p < .05$, ** $p < .01$, *** $p < .001$

Fig. 3 Integrated model of research variables and hypotheses

uct knowledge may facilitate greater openness to acquiring, absorbing, and implementing new structured knowledge during the accelerator process. In this sense, the study empirically demonstrates a compensatory learning mechanism whereby ventures with lower initial knowledge levels benefit more from structured educational interventions and guided learning opportunities. This finding aligns closely with the meta-analysis by Simonsmeier et al. (2022), which shows that while strong prior knowledge predicts higher performance, lower prior knowledge may generate greater learning gains because individuals have more opportunity for cognitive restructuring and knowledge acquisition. Similarly, Chang and Assenova (2026) suggest that entrepreneurs with lower levels of prior knowledge derive greater benefits from the structured components of accelerator programs, as formalized guidance and educational content help compensate for their initial knowledge deficits. The findings of the present study provide empirical support, demonstrating that entrepreneurs with lower levels of prior knowledge benefit to a greater extent from the accelerator's structured program. Together, these findings suggest that accelerator programs may simultaneously reward different mechanisms for different types of ventures: high prior knowledge supports immediate performance and execution, whereas low prior knowledge enhances receptivity to structured learning and adaptive capability. Accordingly, the study contributes to organizational learning and entrepreneurship research by empirically distinguishing between performing well and learning effectively, highlighting that ventures with lower prior core product knowledge may compensate for their initial disadvantages through greater absorptive openness and stronger engagement with accelerator-based learning processes.

3. Intentional pivoting: The findings suggest that mentorship alone is insufficient to improve accelerator outcomes such as Demo Day performance and short-term venture survival. Although accelerator

Table 3 Performance on Demo Day as a mediator of the indirect relationship between meeting with mentors and survival

	B	SE	*t*	*P*	95% CI
		Predicting the mediator Performance on Demo Day			
Mentor Meet	0.05	0.03	1.88	0.066	[−0.00, 0.10]
Pivot	−0.04	0.06	−0.58	0.564	[−0.16, 0.09]
Ment × pivot	0.05	0.02	2.01	0.049	[0.00, 0.10]
		Mentor meeting: simple slope			
Low	−0.01	0.03	−0.34	0.737	[−0.06, 0.05]
Mid	0.05	0.03	1.88	0.066	[−0.00, 0.10]
High	0.11	0.05	2.23	0.023	[0.01, 0.21]
		Predicting the outcome Short-term survival			
Mentor Meet	−0.04	0.09	−0.41	0.685	[−0.21, 0.14]
Demo Day	3.08	0.84	3.68	< 0.001	[1.44, 4.72]
	Indirect moderating effects for various levels of pivot				
Low	−0.03	0.14	–	–	[−0.24, 0.36]
Mid	0.16	0.43	–	–	[0.03, 0.47]
High	0.34	0.86	–	–	[0.08, 0.87]

Mentor Meet meeting with mentors, *Ment* mentors, *Demo Day* performance on demo day

programs emphasize frequent interactions with mentors as a central mechanism for entrepreneurial development (Cohen et al., 2019a, 2019b; Pauwels et al., 2016), the results indicate that the number of mentor meetings alone is not directly associated with stronger Demo Day performance. Similarly, pivoting alone does not appear to improve outcomes and may even reflect reactive or unfocused strategic behavior when undertaken without clear direction. This finding reinforces prior research suggesting that pivoting is inherently ambiguous, as it may represent either adaptive learning or instability and lack of strategic coherence (Hampel et al., 2020; Ries, 2011). Instead, the results demonstrate that pivoting positively moderates the relationship between mentorship and venture outcomes, implying that the value of mentorship emerges when founders engage in intentional and reflective strategic adaptation informed by mentor guidance. In other words, ventures that combine mentor interactions with thoughtful and evidence-based pivots appear better able to translate external advice into improved Demo Day performance and subsequent short-term survival. By contrast, ventures that pivot frantically or opportunistically without integrating mentor input may fail to develop coherent strategic trajectories or executional focus. These findings align with absorptive capacity theory, which emphasizes that the benefits of external knowledge depend not merely on exposure to information, but on the ability to recognize, assimilate, and apply that knowledge effectively (Cohen & Levinthal, 1990; Zahra & George, 2002). The results also resonate with organizational learning theory, which distinguishes productive exploration from excessive or destabilizing experimentation (March, 1991). Within accelerator settings characterized by compressed timelines and high uncertainty, mentor-guided pivoting may serve as a mechanism of purposeful adaptation, enabling ventures to balance learning and strategic coherence while improving both immediate presentation outcomes and early survival prospects.

5.2 Theoretical contributions

5.2.1 Reconciling the ambiguous effects of accelerators

Prior research has produced mixed results regarding whether accelerators improve venture performance (e.g., Hallen et al., 2020; Yu, 2020). This study adds another perspective to the accumulating knowledge on new ventures in accelerators, as part of the effort to clarify these inconsistencies regarding the effect of prior knowledge on new ventures' success in accelerators. The results of our study suggest that accelerators' impact is conditional. It depends on

the interaction between the new venture's pre-entry product knowledge and the motivation to acquire new knowledge through the structured accelerator program.

5.2.2 *Extending absorptive capacity theory to nascent ventures*

Most ACAP studies focus on mature firms (Argote et al., 2020; Camisón & Forés, 2010) with only slim research on new ventures (Larrañeta et al., 2007). By examining early-stage ventures, we extend ACAP theory to contexts of resource scarcity and time constraints.

Our findings reveal a compensatory mechanism: ventures with low pre-entry knowledge gain more from structured learning because they exhibit higher motivation to learn and absorb new knowledge. Conversely, ventures with extensive prior knowledge may experience diminishing returns from a cohort-based structured program (Cohen & Levinthal, 1990). Ventures with high prior knowledge perform well on Demo Day and survive for 2-year post-accelerator, supporting the ACAP theory that prior knowledge leads to success. Whether it leads to new knowledge gain depends on the learning opportunity. Cohort-based structured programs offer less new knowledge to ventures with high prior knowledge. In contrast, they offer opportunities for ventures with limited prior knowledge to compensate for it by acquiring and utilizing new knowledge to strengthen their new venture and succeed.

5.2.3 *Distinguishing potential and realized learning in mentorship*

By differentiating between potential ACAP (PACAP) and realized ACAP (RACAP), we refine the understanding of mentorship in accelerators. Mentorship meetings foster potential learning (acquisition and assimilation), while pivoting represents the realization and exploitation of that learning.

Ventures that translated feedback into intentional pivots achieved the most significant performance gains, illustrating that knowledge implementation, not just acquisition, is the critical link between learning and success.

5.2.4 *Methodological advancement*

The wave-based, multi-cohort design strengthens causal inference by tracing how learning processes unfold over time. This design goes beyond cross-sectional comparisons, offering rare insight into the temporal dynamics of learning and adaptation in accelerator ecosystems.

5.3 Practical implications

For accelerator managers, our findings suggest:

1. Participant selection should consider ventures' pre-entry knowledge levels. Those with limited expertise may benefit most from structured learning, whereas highly experienced teams may require more advanced, customized content.
2. Program design should differentiate between ventures' learning needs. Cohort-based modules are ideal for general knowledge transfer, but high-knowledge ventures may require individualized deep dives.
3. Mentorship management should emphasize actionable learning. Pairing mentors by domain expertise and tracking the extent of pivoting can enhance realized learning outcomes.
4. Evaluation practices, such as Demo Day assessments, serve as valid indicators of future survival, reinforcing their strategic value for identifying scalable ventures.

5.4 Limitations and future research

Although this study offers robust insights, several limitations warrant attention.

First, it examines a single accelerator type: a short, high-intensity, non-equity program. Results may differ in equity-based accelerators that provide financial support or industry specialization. Future studies should test the framework across diverse accelerator models. Given that the first year of post-accelerator survival is considered crucial, as venture performance is primarily realized in the immediate post-acceleration phase (Assenova & Amit, 2024), the present study assessed short-term survival by the ventures' existence for 2-year post-accelerator. This measure fits in with non-funding accelerators that aim to develop new ventures. Since short-term survival may not necessarily predict longer-term

venture performance and outcomes, future studies should investigate venture survival over longer time horizons and for-profit accelerators. Although the present study did not find selection bias parameters, such as gender (Dams et al., 2022), future research can further examine other selection biases, such as education, age, and work experience. Preliminary core product knowledge of new ventures could be further assessed beyond self-report.

Finally, future work could explore team-level cognition and feedback orientation (Linderbaum & Levy, 2010) as micro foundations of absorptive capacity in nascent ventures.

5.5 Conclusion

The present study advances the understanding of accelerators as learning environments that interact with new ventures' prior knowledge to affect their performance upon accelerator graduation and short-term survival through absorptive capacity for new knowledge acquisition and realization through mentorship and pivoting. This study demonstrates that one size does not fit all. The fit depends on the interaction between the new venture's prior knowledge and capabilities and the accelerator type and structure. By introducing ACAP theory to the entrepreneurship ecosystem, we highlighted the compensatory effect for new ventures with low prior knowledge in non-profit accelerators that offer structured, cohort-based programs. Furthermore, ACAP theory contributed to understanding the impact of mentorship on new venture success. Our findings showed that PACAP, representing the acquisition and assimilation of new knowledge from mentors, influences performance only when it is leveraged through intentional pivots.

Author contributions SLK designed the research, collected the data, analyzed the data, wrote the paper, reviewed the paper ME designed the research, reviewed the data analysis, wrote the paper, reviewed the paper.

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Data availability Our data will be available upon request.

Declarations

Ethics approval and consent to participate Ethics and consent to participate declarations: IRB

Competing interests The authors declare no competing interests.

Appendix A

Table 4 The accelerator curriculum

Session No	Topic
1st	Opening
2nd	Design thinking
3rd	Business model canvas + exercise
4th	Customer segmentation–value proposition canvas1 (VPC1)
5th	Lab time*: building questionnaires
6th	Market research + ecosystem analysis + product market fit
7th	Minimum viable product MVP + VPC2
8th	MVP planning
9th	Pitching and storytelling
10th	Lab time*: pitching practice + MVP
11th	"Shark Tank" ("dry run" of pitching)
12th	Marketing + advertising
13th	Gameplan1: the history of future
14th	Gameplan2: building a budget/financial plan
15th	Gameplan3: business plan + SWOT
16th	Business consulting
17th	Lab time*: pitching
18th	Demo Day – "dry run"
19th	Demo Day

Lab time* means working on a topic during the time at the accelerator

Weekly plan of accelerator program handed to the new ventures at the beginning of the batch

Appendix B

ACAP Questionnaire – Flatten et al. (2011)

Seven-point Likert scale ranging from 1 ("strongly disagree") to 7 ("strongly agree")

Acquisition

Please specify to what extent your venture uses external resources to obtain information (e.g., the accelerator's offerings: training, mentorship, networking, personal networks, consultants, seminars,

internet database, professional journals, academic publications, market research, regulations and laws concerning environment/technology/health…).

Acquisition 1The search for relevant information concerning our industry is everyday business in our venture.

Acquisition 2This venture encourages its members to use information sources within our industry.

Acquisition 3 In this venture, the expectation is that the members gather information beyond our industry.

Assimilation

Please rate to what extent the following statements fit the communication structure in your venture:

Assimilate 1In our venture, ideas and concepts are communicated cross-expertise.

Assimilate 2 This venture emphasizes cross-expertise support to solve problems.

Assimilate 3 In our venture, there is a quick information flow across expertise.

Assimilate 4 Our venture encourages periodical cross-expertise meetings to share new developments, problems, and achievements.

Transform

Please specify to what extent the following statements fit the knowledge processing in your company:

Transform 1Our venture has the ability to structure and to use collected knowledge.

Transform 2Our venture is used to absorb new knowledge as well as to prepare it for further purposes and to make it available.

Transform 3Members of this new venture successfully link existing knowledge with new insights.

Transform 4This venture applies new knowledge into practical work.

Exploit

Please specify to what extent the following statements fit the commercial exploitation of new knowledge in your venture (NB: Please think about all venture divisions such as R&D, production, marketing, and accounting):

Exploit 1This venture supports the development of prototypes.

Exploit 2This venture regularly reconsiders technologies and adapts them to create new knowledge.

Exploit 3This venture capable of working more effectively by adopting new technologies.

Core product knowledge questionnaire items:

Please mark to what extent the following content areas and expertise does your venture currently holds internally. Ranging from 1-strongly disagree to 5-strongly agree.

Our venture holds internally:

1. Relevant technological expertise
2. Relevant production expertise

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